

Programming Fundamentals – COC1071

Lab Plan – 04



Learning Objective:

- Repetition

- While loop
- Do-while loop

- 1) Write a program that counts the number of digits in a number entered by the user.
- 2) Write a program that calculates the sum of all numbers in a range specified by the user.
- 3) Write a program that prints all numbers between 1 and 100 that are divisible by both 5 and 7.
- 4) Write a program that accepts 10 numbers from the user and calculates the positive numbers and negative numbers separately.
- 5) Write a program that calculates the sum of a 5-digit number entered by the user.
- 6) Write a program that checks if a number entered by the user, is a palindrome or not.
- 7) Write a program that converts a 3-digit entered by the user into its corresponding word.
- 8) Write a program that checks if a number entered by the user, is a prime number. A prime number is a number greater than 1 that has no divisors other than 1 and itself.
- 9) Write a program that generates the first N numbers in the Fibonacci sequence. The Fibonacci sequence is: 0, 1, 1, 2, 3, 5, 8, ... Input the number from the user and generate the sequence.
- 10) Write a program that calculates the GCD of two numbers entered by the user. Use the Euclidean algorithm, where GCD of a and b is found by repeatedly replacing a by b and b by $a \% b$ until $b=0$.